

Embedded Artificial Intelligence in Battery Management Systems: Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation

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Abstract

Online battery management requires accurate state-of-charge (SOC) and state-of-health (SOH) estimates within strict memory and timing budgets. To address these constraints, this work investigates how stateful long short-term memory (LSTM) networks with multi-layer perceptron (MLP) heads can be optimised for embedded artificial intelligence (AI), using structured pruning and recurrent-weight quantization to reduce memory and computational costs while retaining estimation accuracy on an STM32H753ZI microcontroller. Using the open *Lithium Iron Phosphate (LFP) Cycle Ageing Dataset*, full-precision models provide the accuracy reference against which the compressed variants are evaluated. Long-horizon replay yields mean absolute error (MAE) values of 2.3–2.8 percentage points for SOC and 0.9–1.5 for SOH, showing that compression largely preserves accuracy. Hardware-in-the-loop benchmarks show that the evaluated 30 % pruning configuration reduces flash by 40.9 % and 45.5 % and inference time by 42.9 % and 44.0 % for the SOC and SOH models, respectively. The analytical complexity model links these savings to pruning fraction and network dimensions. Quantization further reduces flash by 50.2 % and 58.8 %, but raises runtime in the mixed-precision implementation. Together, the study provides a reproducible reference for resource-efficient embedded battery-state estimation.

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CRedit authorship contribution statement

Florian Rzepka: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing–original draft preparation, writing–review and editing, visualization. **Julia Kowal:** Resources, supervision, project administration, funding acquisition, writing–review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT, DeepL, and Grammarly to improve the linguistic quality and style of the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.